

Ch 11: Electricity

Part 1 of 1 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION A - INTRODUCTION TO ELECTRICITY, ELECTRIC CHARGE AND ELECTRIC CURRENT

1. Electricity in Daily Life

- Electricity is a form of energy that makes our lives comfortable, convenient and productive.
- It is used in lighting, heating, cooking, communication, transport, entertainment, industries and in almost every modern device.
- From switching on a bulb to charging a phone—electricity is everywhere in daily life.

2. Electric Charge

- Matter is made up of tiny particles called atoms.
- Atoms contain positively charged particles (protons) and negatively charged particles (electrons).
- An object becomes electrically charged when there is an imbalance in the number of electrons.

Positive and Negative Charge

- If an object loses **electrons**, it has more protons than electrons and becomes **positively charged**.
- If an object **gains electrons**, it has more electrons than protons and becomes **negatively charged**.
- Like charges repel each other; unlike charges attract each other.

3. Static Electricity and Current Electricity

- Static Electricity** – Charge at rest on an object (e.g., comb rubbed with hair, balloon rubbed on wool).
- Current Electricity** – Continuous flow of electric charge through a conductor (e.g., current in household wiring).

4. Electric Current

- Electric current is the rate of flow of electric charge through a conductor.
- It is a scalar quantity.

Current as Flow of Charge

- If a charge Q coulomb flows through a conductor in time t second, the current I is given by:

$$I = \frac{Q}{t}$$

where, I = current (A), Q = charge (C), t = time (s)

- SI unit of current is AMPERE (A).**
- 1 ampere = 1 coulomb of charge flows per second through the cross-section of a conductor.**

Common Units of Current

Unit	Symbol	Relation	Meaning
Ampere	A	1 A = 1 C/s	One coulomb per second
Milliampere	mA	1 mA = 10^{-3} A	One thousandth of an ampere
Microampere	μ A	1 μ A = 10^{-6} A	One millionth of an ampere

Time and Charge

- Time unit: second (s)
- Charge unit: coulomb (C)
- 1 C is the amount of charge carried by a current of 1 A in 1 s.

Examples of Current

- 60 C of charge in 1 minute $\rightarrow I = 60/60 = 1$ A
- 0.5 C of charge in 2 s $\rightarrow I = 0.5/2 = 0.25$ A
- 180 mC in 30 s $\rightarrow I = 0.18/30 = 6$ mA

QUICK RECALL

- Like charges repel, unlike charges attract.
- Current = Flow of charge.
- $I = Q/t$; Unit of current: ampere.
- Closed circuit needed for current.
- Cell $\rightarrow$ Battery $\rightarrow$ Circuit $\rightarrow$ Appliance (e.g., bulb).

KEY IDEA

Electric current is the continuous, ordered flow of electric charge. To make current flow, we need a source (cell or battery) to push the charges and a closed conducting path for the charges to move through. Without a complete path, the flow stops.

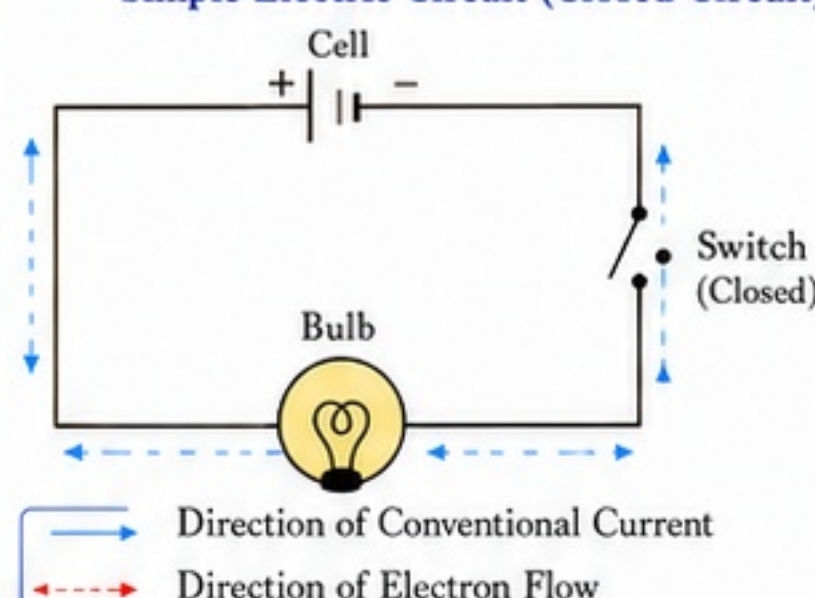
5. Sources of Electric Current

- Electric current is produced by sources that maintain a potential difference (voltage) between two points.
- Common sources:
 - Cell:** A single cell produces a small potential difference.
 - Battery:** Two or more cells connected together provide a higher potential difference and can supply current for longer.

6. Simple Electric Circuit

- A circuit is a complete and closed conducting path through which electric current flows.
- A circuit must be closed; otherwise, current will not flow.

Simple Electric Circuit (Closed Circuit)



Why Closed Circuit

- Current flows only when the circuit forms a complete loop.
- If the circuit is open at any point, the flow of charge stops and the bulb does not glow.

Open Circuit vs Closed Circuit

Open Circuit (Switch Open)	Closed Circuit (Switch Closed)
- Path is incomplete. - No current flows. - Bulb does not glow. - Appliances do not work.	- Path is complete. - Current flows. - Bulb glows. - Appliances work.

7. Essential Elements of an Electric Circuit

- Cell** – Provides potential difference (voltage).
- Battery** – Combination of two or more cells for higher voltage.
- Bulb (Lamp)** – Converts electrical energy into light energy.
- Switch (Key)** – Makes or breaks the circuit.
- Connecting Wire** – Provides low-resistance path for current.
- Resistor** – Offers resistance to control the flow of current.

8. Basic Circuit Symbols

Component	Symbol	Meaning
Cell		Single cell
Battery		Two or more cells
Bulb (Lamp)		Glowing lamp
Switch (Open)		Circuit open
Switch (Closed)		Circuit closed
Resistor		Resistor
Connecting wire		Conductor

EXAM TIPS

- Remember the formula $I = Q/t$ and units (A, mA, μ A, C, s).
- Learn circuit symbols well; they are asked in diagrams.
- Mark conventional current from + to -; electron flow is opposite.
- Always check if the circuit is closed for current to flow.

CENTUM FOCUS

- Draw neat, labeled circuit diagrams.
- State units correctly in answers.
- Write definitions in your own words.
- Give 2–3 correct examples where asked.
- Practice numerical problems on $I = Q/t$ regularly.

Ch 11: Electricity

SECTION B - ELECTRIC POTENTIAL, POTENTIAL DIFFERENCE AND MEASURING CURRENT

1. Electric work in moving charge

- When a charge q is moved from one point to another in an electric field, the electric force does work on it.
- If W joules of work are done in moving a charge q coulombs, that work is stored as electric potential energy.
- Work is needed to move a positive charge from lower potential to higher potential.

2. Electric potential (V)

- Electric potential at a point is the work done per unit positive charge in bringing the charge from infinity to that point without any acceleration.
- Formula: $V = \frac{W}{Q}$
- Where, W = work done (joule), Q = charge (coulomb)
- SI unit: volt (V) [1 volt = 1 joule/1 coulomb]

Meaning of 1 volt

1 volt is the potential when 1 joule of work is done in bringing 1 coulomb of positive charge from infinity to that point.

3. What does a cell do?

- A cell maintains a potential difference (EMF) between its two terminals.
- It does work on charges inside the circuit and moves them from its negative terminal to its positive terminal.
- It is the source of electric energy in a circuit.

4. Potential difference between two points

- Potential difference (V_{AB}) between two points A and B is the work done per unit charge in moving a charge from A to B.
- $V_{AB} = \frac{W_{AB}}{Q}$
- Unit: volt (V)
- It depends on the electric field and path between the two points.

Why potential difference is required for current flow?

- Charges flow only when there is a difference in electric potential.
- The higher potential (positive terminal of cell) pushes charges and the lower potential (negative terminal) pulls them.
- Without potential difference, charges will not flow and no current will be produced.

5. Electric current vs Potential difference

Feature	Electric Current (I)	Potential Difference (V)
Meaning	Rate of flow of electric charge through a conductor.	Work done per unit charge between two points.
Formula	$I = \frac{Q}{t}$	$V = \frac{W}{Q}$
Unit	Ampere (A)	Volt (V)
Depends on	Charge and time.	Work done and charge.
Role in circuit	Causes heating, magnetism, etc.	Causes current to flow.
Nature	Flow (rate quantity)	Driving force (potential energy difference)

6. Electromotive Force (EMF)

- The EMF of a cell is the potential difference between its two terminals when no current is drawn from it.
- It is the maximum potential difference provided by the cell.
- Unit: volt (V)
- EMF is denoted by E .

Key idea

EMF is the energy given by the cell to each coulomb of charge to move it from the negative to the positive terminal.

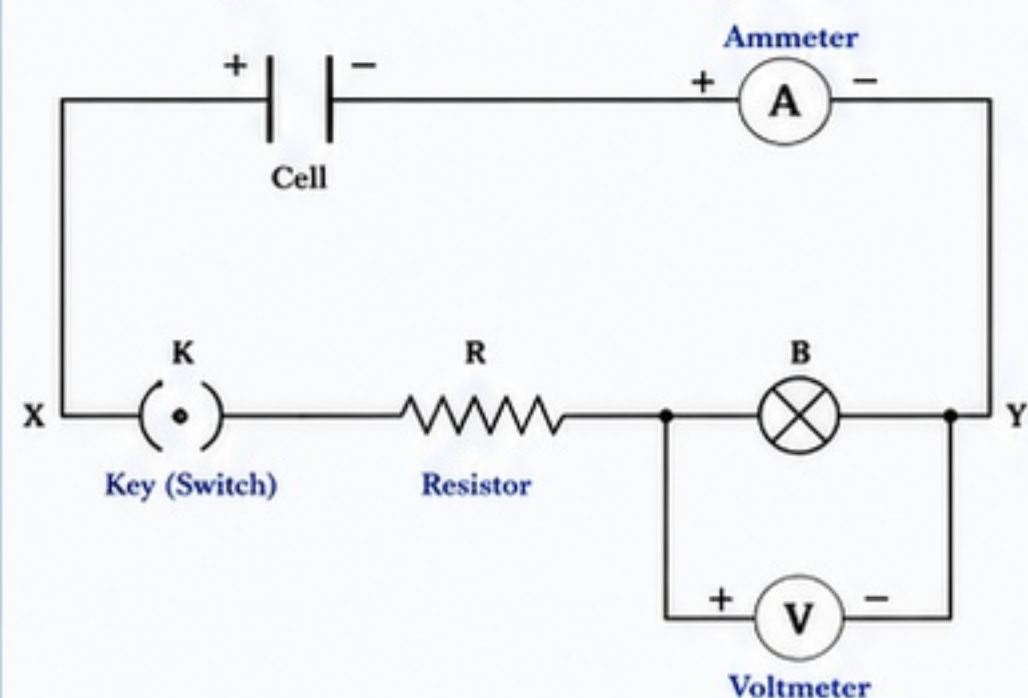
7. Measuring current using ammeter

- An ammeter is used to measure electric current in a circuit.
- It must be connected in series with the component whose current is to be measured.
- Key rule: Ammeter has very low resistance. In series connection, it does not change the current of the circuit.
- The positive (+) terminal of ammeter is connected towards the positive terminal of the cell.

8. Measuring potential difference using voltmeter

- A voltmeter is used to measure potential difference between two points.
- It must be connected in parallel across the component or points.
- Key rule: Voltmeter has very high resistance. In parallel connection, it does not draw significant current.
- The positive (+) terminal of voltmeter is connected to the point of higher potential.

9. Ideal placement of circuit components



Component	Symbol	How to connect	Purpose
Cell	$+ -$	—	Provides potential difference (EMF) in the circuit.
Key (Switch)	K (K)	Series	To make or break the circuit.
Resistor	zigzag	Series	To limit current and oppose its flow.
Bulb (Lamp)	$\otimes$	Series	To glow when current flows.
Ammeter	A	Series	To measure current in the circuit.
Voltmeter	V	Parallel	To measure potential difference across a component.

10. Electric circuit safety – basic rules

- Always use proper insulation: Wires should have good insulation to prevent electric shock.
- Do not touch live wires or open terminals of a cell.
- Use correct connections: Ammeter in series, voltmeter in parallel.
- Do not short circuit a cell: Directly connecting the terminals can cause high current and heat.
- Use components within their rated limits.
- Switch off the circuit before making or changing connections.



QUICK RECALL

- $V = \frac{W}{Q}$ | $I = \frac{Q}{t}$
- 1 volt = 1 joule per coulomb.
- Cell provides potential difference.
- Current flows only when there is potential difference.
- Ammeter → series | Voltmeter → parallel.



KEY IDEA

Potential difference is the driving force that makes charges move. The cell maintains this difference. Current is the flow of charges due to this driving force. Measure current with an ammeter in series and potential difference with a voltmeter in parallel.



EXAM TIPS

- Write formulas with symbols clearly (V , I , W , Q , t).
- Always mention units.
- Draw neat and labelled circuits.
- State the rule: Ammeter in series, voltmeter in parallel.
- Add key points like "Why" or "What does" in answers.



CENTUM FOCUS

- Learn definitions word-to-word.
- Draw the circuit from memory.
- Compare current and potential difference in tabular form.
- Solve numericals on $V = W/Q$ and $I = Q/t$.
- Practise NCERT in-text questions.

Ch 11: Electricity

SECTION C - OHM'S LAW, RESISTANCE AND V-I RELATION

1. Ohm's Law – Statement

Ohm's law states that, at **constant temperature**, the **current** flowing through a conductor is directly proportional to the potential difference across its ends.

2. Condition

- Ohm's law holds good only when the temperature of the conductor remains constant. If temperature changes, the resistance may change and the law may not be valid.

3. Mathematical Relation

$$V = IR$$

V – Potential difference (volt, V)

I – Current (ampere, A)

R – Resistance (ohm, Ω)

4. Meaning of Resistance

- Resistance is the property of a conductor that opposes the flow of current.

5. SI Unit

- The SI unit of resistance is **ohm** (symbol Ω).

6. Meaning of 1 Ohm

- A conductor has a resistance of 1 ohm if a potential difference of 1 volt across its ends produces a current of 1 ampere through it.

7. Why Resistance Opposes Current?

- When electrons flow through a conductor, they collide with the atoms and ions present in it. These collisions offer opposition to their motion. This opposition is called resistance.

8. Relation Between V , I and R

- From $V = IR \Rightarrow I = \frac{V}{R}$ and $R = \frac{V}{I}$
- For a given conductor (at constant temperature), R is constant. Therefore, $V \propto I$.

9. Practical Meaning

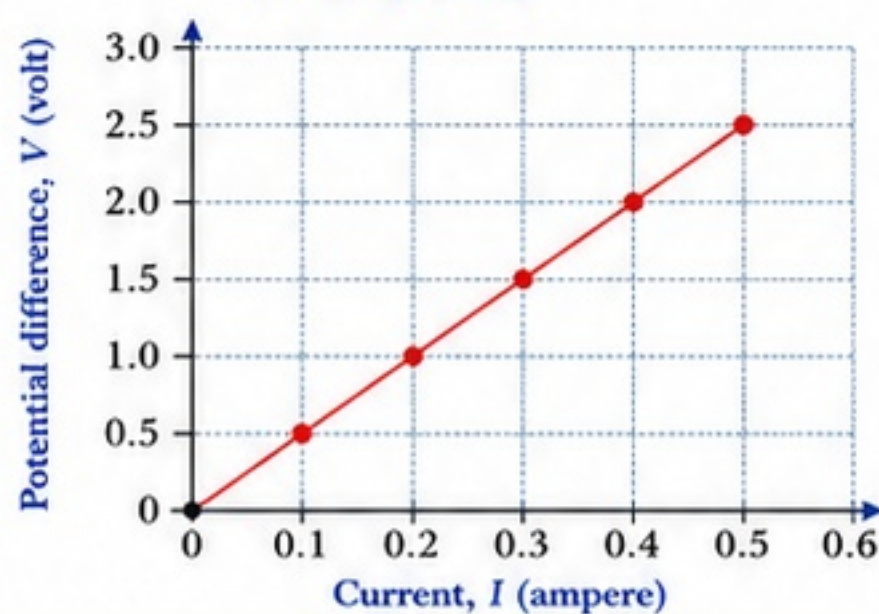
- High Resistance** – Less current flows for the same voltage.
- Low Resistance** – More current flows for the same voltage.

11. OBSERVATIONS AND CALCULATIONS

S. No.	Potential Difference V (volt)	Current I (ampere)	V/I (ohm)
1	0.5	0.10	5.0
2	1.0	0.20	5.0
3	1.5	0.30	5.0
4	2.0	0.40	5.0
5	2.5	0.50	5.0

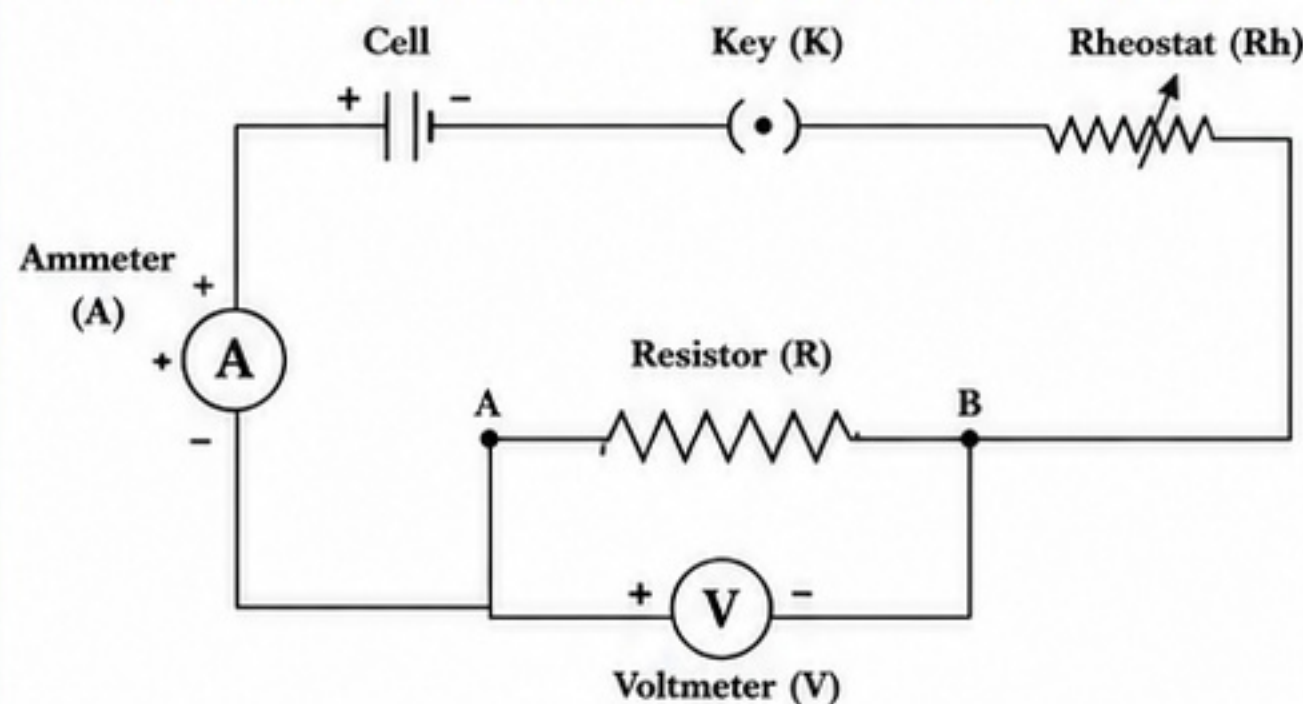
Observation: For all readings, V/I is constant. Hence, resistance $R = 5.0 \Omega$ is constant.

12. V-I GRAPH



The V - I graph is a straight line passing through the origin. This shows that $V \propto I$ and Ohm's law is valid for the conductor.

10. EXPERIMENTAL SETUP TO VERIFY OHM'S LAW



- Make the circuit as shown.
- Keep the rheostat at maximum resistance. Close the key.
- Note the current (I) in ammeter and potential difference (V) in voltmeter.
- Change the rheostat to decrease the resistance gradually.
- Note different values of V and I .
- Plot a graph between V (y-axis) and I (x-axis).

13. FORMULAS AND UNIT RELATIONS

- $V = IR$ (Ohm's law)
- $I = \frac{V}{R}$ (Current)
- $R = \frac{V}{I}$ (Resistance)
- SI unit of V – volt (V)
- SI unit of I – ampere (A)
- SI unit of R – ohm (Ω)
- $1 \text{ ohm} = 1 \frac{\text{volt}}{\text{ampere}}$
- $1 \text{ kilo-ohm (k } \Omega) = 10^3 \Omega$
- $1 \text{ mega-ohm (M } \Omega) = 10^6 \Omega$

14. NON-OHMIC CONDUCTORS (NOTE)

- Ohm's law is not followed by all materials. In some materials (e.g., filament of a lamp, thermistor), the V - I graph is not a straight line through the origin.



QUICK RECALL

- $V \propto I$ at constant temperature.
- $V = IR$
- Unit of resistance – ohm (Ω).
- $1 \Omega \rightarrow 1 \text{ V}$ produces 1 A.
- Resistance opposes current.



KEY IDEA

Resistance is the reason why a given voltage does not produce unlimited current. It depends on the material, length, area of cross-section and temperature. For a conductor at constant temperature, V - I relation is linear.



EXAM TIPS

- Always mention "at constant temperature".
- Write formula with correct symbols and units.
- Use proper units in answers.
- Draw V - I graph neatly with labels and straight line through origin.



CENTUM FOCUS

- Memorise formulas and unit relations.
- Practise numerical problems daily.
- Draw neat circuit diagrams.
- Write observations and conclusion clearly as in practical.



Ch 11: Electricity

SECTION D - FACTORS AFFECTING RESISTANCE AND RESISTIVITY

1. FACTORS AFFECTING RESISTANCE OF A CONDUCTOR

The resistance (R) of a conductor depends on four main factors:

(1) Length (l) of the conductor

- Longer the conductor, greater the resistance.

Directly proportional: $R \propto l$

(2) Cross-sectional area (A) of the conductor

- Thicker the conductor, lesser the resistance.

Inversely proportional: $R \propto \frac{1}{A}$

(3) Nature of the material (depends on resistivity ρ)

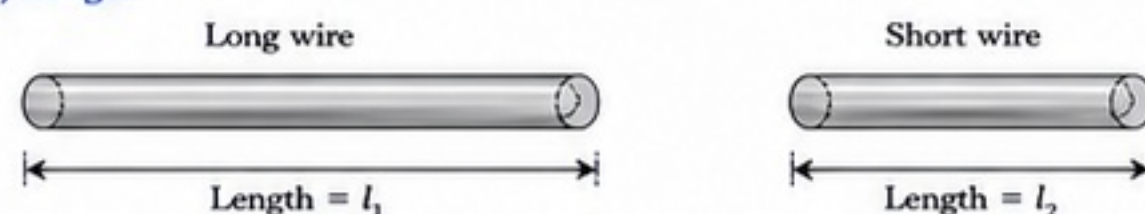
- Different materials offer different resistance.

(4) Temperature (T) of the conductor

- In metals, resistance increases with rise in temperature.
- In some alloys (e.g., nichrome), resistance is almost unaffected by temperature.

2. EFFECT OF LENGTH AND CROSS-SECTIONAL AREA

(A) Length



If $l_1 > l_2$, then $R_1 > R_2$ (Longer wire has more resistance)

(B) Cross-sectional area



If $A_1 > A_2$, then $R_1 < R_2$ (Thicker wire has less resistance)

3. COMBINED RELATION AND RESISTIVITY

- Combining the effects of length and area, the resistance of a conductor is given by:

$$R = \rho \frac{l}{A}$$

Formula Box

- $R \propto l$
- $R \propto \frac{1}{A}$
- $R \propto \rho$
- Combined:
 $R = \rho \frac{l}{A}$

Where,

R = resistance of the conductor (ohm, Ω)

l = length of the conductor (metre, m)

A = cross-sectional area (square metre, m^2)

ρ = resistivity of the material (ohm metre, Ωm)

- Resistivity (ρ)** is a measure of how strongly a material opposes the flow of electric current.
- It is independent of the dimensions of the conductor.
- SI unit of resistivity is **ohm metre (Ωm)**.
- Lower the resistivity, better the conductor.

4. RESISTANCE VS RESISTIVITY

Basis	Resistance (R)	Resistivity (ρ)
Meaning	Opposition offered by a conductor to the flow of current.	Intrinsic property of a material that indicates its ability to oppose current.
Depends on	Length, area, material and temperature.	Nature of material and temperature only.
Depends on size and shape	Yes	No
SI unit	Ohm (Ω)	Ohm metre (Ωm)
Formula	$R = \rho \frac{l}{A}$	$\rho = \frac{RA}{l}$
Nature	Property of a conductor.	Property of a material.

5. CONDUCTORS, INSULATORS AND RESISTIVE ALLOYS

	Good Conductors (Low Resistivity)	Poor Conductors / Insulators (High Resistivity)	Resistive Alloys (High Resistivity, Stable)
Examples	Copper, Aluminium, Silver, Gold	Rubber, Plastic, Glass, Wood, Dry air, Porcelain	Nichrome, Manganin, Constantan
Resistivity	Very low	Very high	High
Use	Electrical wires, cables	Insulation, safety	Heating elements, rheostats, resistance coils

6. WHY DIFFERENT MATERIALS ARE USED?

- Copper and aluminium are used for electrical wires because they have very low resistivity, are ductile, and can be drawn into thin wires easily.
- Nichrome** (an alloy of nickel and chromium) has high resistivity and does not oxidise easily. It can be heated to red-hot temperature without melting, so it is used in heating elements (e.g., heaters, toasters, electric ovens).
- Manganin** is used in precision resistors because its resistivity is nearly independent of temperature.

7. EFFECT OF TEMPERATURE ON RESISTANCE

- In metals (e.g., copper, aluminium): As temperature increases, atoms vibrate more, causing more collisions with electrons, so resistance increases.
- In alloys like nichrome and manganin: Resistance remains almost constant even when temperature changes.

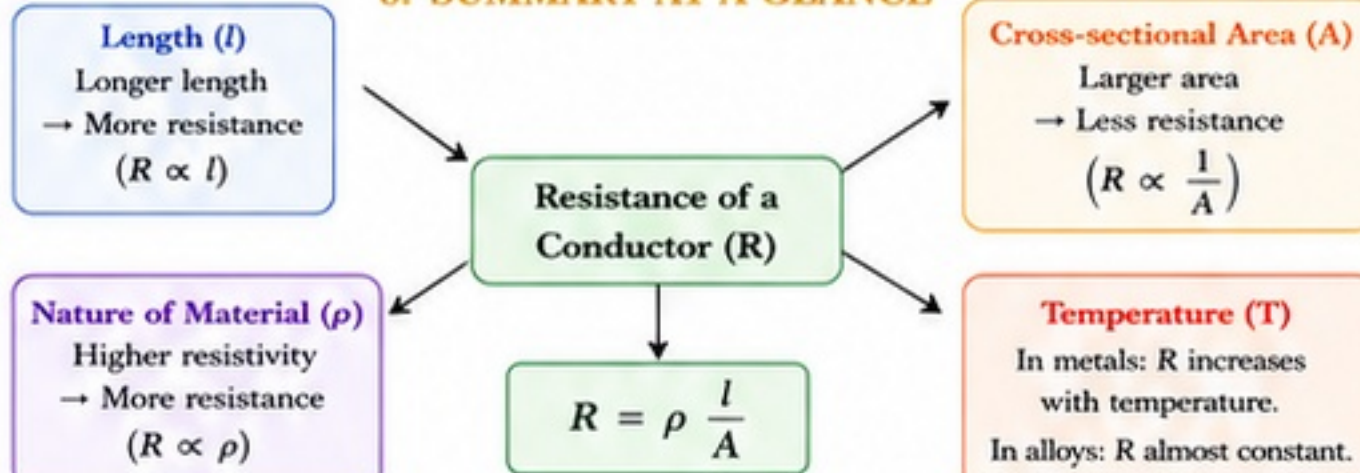
Metals

R increases with temperature
(Positive temperature coefficient)

Alloys (e.g., Nichrome, Manganin)

R almost constant with temperature
(Nearly zero temperature coefficient)

8. SUMMARY AT A GLANCE



QUICK RECALL

- $R \propto l$ (Longer → more R)
- $R \propto \frac{1}{A}$ (Thicker → less R)
- $R \propto \rho$ (Greater ρ → more R)
- ρ = property of material.
- SI unit of ρ is Ωm .



KEY IDEA

- Resistance tells 'how much', resistivity tells 'how strongly' a material resists the flow of current.
- Low ρ = good conductor; High ρ = poor conductor or insulator.
- Alloys have high ρ and suitable temperature properties for heating and control.



EXAM TIPS

- Remember the combined formula: $R = \rho \frac{l}{A}$.
- Learn units: R (Ω), ρ (Ωm), l (m), A (m^2).
- Write difference between R and ρ clearly.
- Use the concept map for quick recall.



CENTUM FOCUS

- Practise numericals using $R = \rho \frac{l}{A}$.
- Compare materials using resistivity values from tables in the book.
- Relate concepts to real-life examples like wires, heaters, and insulators.

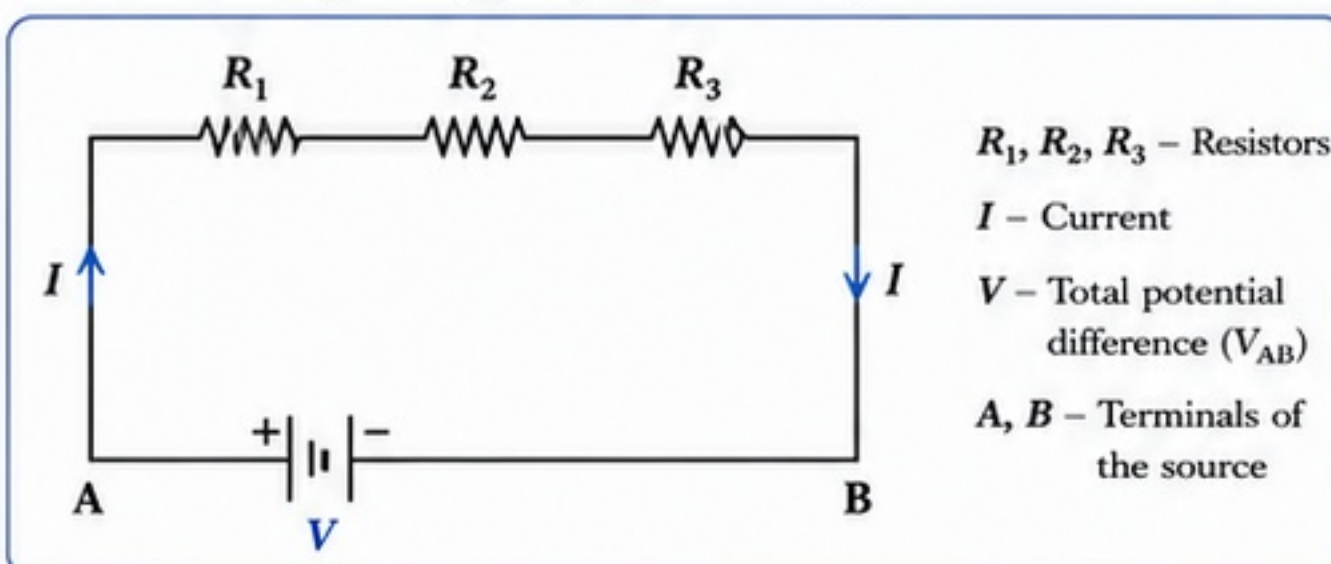
Ch 11: Electricity

SECTION E - COMBINATION OF RESISTORS IN SERIES

1. What is series combination?

- Resistors are said to be in **series** when they are connected end-to-end so that the same current flows through each resistor one after another.
- There is only one path for current to flow.

2. Circuit diagram (R_1, R_2, R_3 in series)



3. Key points in series combination

- Same current:** The current through each resistor is the same.
- Voltage division:** The total potential difference across the combination equals the sum of potential differences across individual resistors.

$$V = V_1 + V_2 + V_3$$

where V_1, V_2, V_3 are potential differences across R_1, R_2, R_3 respectively.

4. Derivation of equivalent resistance

- Let current through each resistor be I .
- Potential difference across each resistor:

$$V_1 = IR_1, \quad V_2 = IR_2, \quad V_3 = IR_3$$

- Total potential difference:

$$V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3 = I(R_1 + R_2 + R_3)$$

- By Ohm's law for the total combination: $V = IR_s$

Therefore,

$$R_s = R_1 + R_2 + R_3$$

DERIVATION FLOW

$$V_1 = IR_1, V_2 = IR_2, V_3 = IR_3$$

$$V = V_1 + V_2 + V_3$$

$$= I(R_1 + R_2 + R_3)$$

$$V = IR_s$$

$$R_s = R_1 + R_2 + R_3$$

5. Why equivalent resistance increases in series?

- In series, resistors add to the path of current.
- More opposition is offered to the flow of current.
- Therefore, the total resistance (R_s) is greater than any individual resistor.

6. Effect on current

As equivalent resistance increases, the current decreases (for a given voltage).

$$I = \frac{V}{R_s}$$

7. Solved Example

Three resistors of $2\ \Omega$, $3\ \Omega$ and $5\ \Omega$ are connected in series across a 12 V battery. Find:

- (i) Equivalent resistance (ii) Current in the circuit (iii) Voltage across each resistor.

Solution:

(i) $R_s = R_1 + R_2 + R_3 = 2 + 3 + 5 = 10\ \Omega$

(ii) $I = \frac{V}{R_s} = \frac{12}{10} = 1.2\text{ A}$

(iii) Voltage across each resistor:

- $V_1 = IR_1 = 1.2 \times 2 = 2.4\text{ V}$

- $V_2 = IR_2 = 1.2 \times 3 = 3.6\text{ V}$

- $V_3 = IR_3 = 1.2 \times 5 = 6.0\text{ V}$

Check: $V_1 + V_2 + V_3 = 2.4 + 3.6 + 6.0 = 12\text{ V}$ ✓

8. Comparison: Series combination at a glance

Aspect	In Series	
Path of current	Only one path	
Current (I)	Same through all resistors	$I_1 = I_2 = I_3 = I$
Potential difference (V)	Divided across resistors	$V = V_1 + V_2 + V_3$
Equivalent resistance (R_s)	Sum of resistances	$R_s = R_1 + R_2 + R_3$
Effect on current	Increases R_s , decreases I	$I = \frac{V}{R_s}$
If one resistor breaks	Circuit is open, current stops through all	No current flows

9. Practical uses of series circuits

- Electric bulbs in old-style decorative strings.
- Christmas lights and decorative lighting.
- Voltage dividers in electronic circuits.
- Measuring devices (e.g., ammeter is connected in series).

10. Disadvantage of series connection

- If one component (e.g., bulb or resistor) breaks, the circuit opens.
- Current stops flowing through all components.
- Other devices in the circuit will not work.

11. Formula Summary

1. Same current: $I_1 = I_2 = I_3 = I$

2. Total voltage: $V = V_1 + V_2 + V_3$

3. Equivalent resistance: $R_s = R_1 + R_2 + R_3$

4. Ohm's law: $V = IR_s$, $I = \frac{V}{R_s}$



QUICK RECALL

- Series means end-to-end.
- Same current everywhere.
- Voltages add up.
- Resistances add up.
- If one breaks, all stop!



KEY IDEA

In series, resistors share the same current but divide the voltage. Each resistor drops some voltage proportional to its resistance. More resistors in series mean more total resistance and less current for the same voltage.



EXAM TIPS

- Always write I same for all in series.
- Add resistances to get R_s .
- Use $V = IR$ for each resistor to find voltage drops.
- Check: Sum of drops = Total V .



CENTUM FOCUS

- Practise 2–3 numerical problems with different values.
- Draw neat circuit diagrams.
- State formulas before solving.
- Write the unit with every answer.



Revise smart, draw diagrams neatly, and score full marks.



Ch 11: Electricity

SECTION F - COMBINATION OF RESISTORS IN PARALLEL

1. What is parallel combination of resistors?

- When two or more resistors are connected across the same two points, such that each resistor forms a separate branch, the combination is called parallel combination.
- Each resistor has the same potential difference across its ends, while the current divides among the branches.

2. Key features

- Same potential difference (V)** across each branch.
- Total current (I)** supplied by the battery equals the sum of currents in all branches.
- Equivalent resistance (R_p)** is less than the smallest branch resistance.
- If one branch is removed, others continue to work.

3. Mathematical derivation of equivalent resistance

Derivation

- Let the potential difference across the combination be V . Currents in the three branches are I_1 , I_2 and I_3 .
- By Ohm's law,

$$I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}, \quad I_3 = \frac{V}{R_3}$$

- Total current, $I = I_1 + I_2 + I_3$

$$I = V \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

- Also, $I = \frac{V}{R_p}$, where R_p is the equivalent resistance.

$$\frac{V}{R_p} = V \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

- Cancelling V ,

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

7. Solved Example

Three resistors of $6\ \Omega$, $3\ \Omega$ and $2\ \Omega$ are connected in parallel across a 12 V battery. Find: (i) Equivalent resistance, (ii) Total current, (iii) Current through each resistor.

Solution:

(i) Using $\frac{1}{R_p} = \frac{1}{6} + \frac{1}{3} + \frac{1}{2} = \frac{1}{6} + \frac{2}{6} + \frac{3}{6} = \frac{6}{6} = 1$

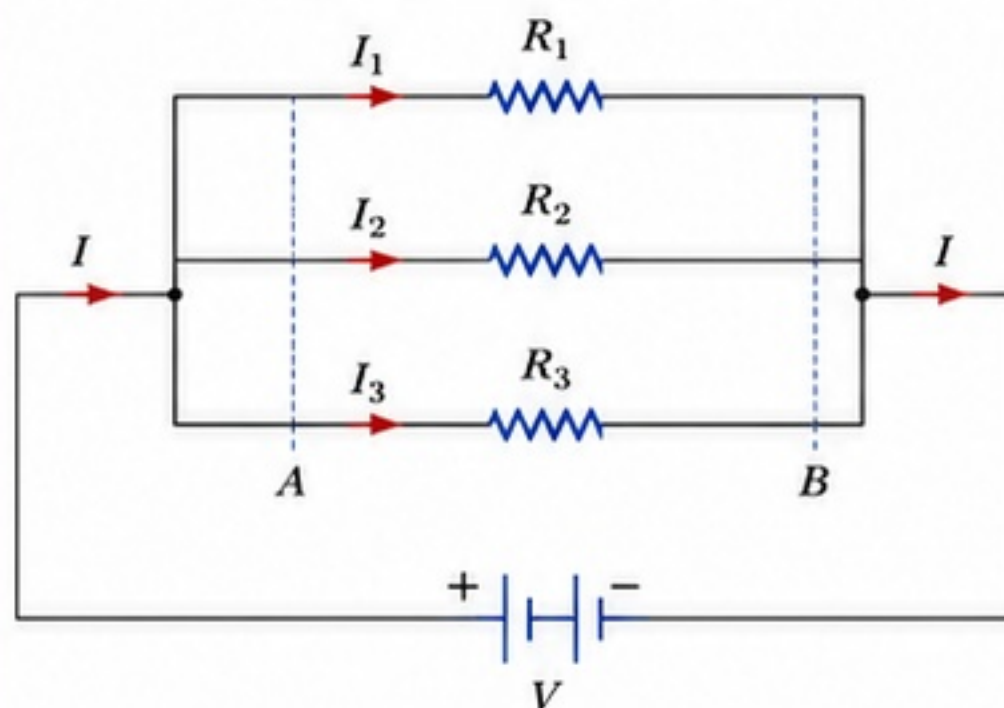
$$R_p = 1\ \Omega$$

(ii) Total current, $I = \frac{V}{R_p} = \frac{12}{1} = 12\text{ A}$

(iii) $I_1 = \frac{V}{R_1} = \frac{12}{6} = 2\text{ A}$; $I_2 = \frac{12}{3} = 4\text{ A}$; $I_3 = \frac{12}{2} = 6\text{ A}$

Check: $I = I_1 + I_2 + I_3 = 2 + 4 + 6 = 12\text{ A}$ ✓

CIRCUIT DIAGRAM – RESISTORS IN PARALLEL



What this shows

- R_1 , R_2 and R_3 are connected in parallel between points A and B.
- Same potential difference V across each resistor.
- I is the total current supplied by the battery.
- $I = I_1 + I_2 + I_3$

4. Why is equivalent resistance smaller?

- In parallel, adding more branches gives more paths for current to flow.
- Total current increases for the same voltage, so equivalent resistance decreases.

R_p is always less than the smallest of R_1 , R_2 or R_3 .

5. Current distribution in branches

- If resistances are different, currents divide inversely proportional to resistance.
- The smaller the resistance, the larger the current in that branch.

$$\frac{I_1}{I_2} = \frac{R_2}{R_1}, \quad \frac{I_2}{I_3} = \frac{R_3}{R_2}, \quad \frac{I_1}{I_3} = \frac{R_3}{R_1}$$

6. Why parallel combination is used in homes and practical circuits

- Each appliance gets the full potential difference of the supply (e.g., 230 V AC).
- Appliances operate independently – one can be switched off without affecting others.
- Failure of one appliance does not stop others from working.
- Total current drawn is the sum of currents required by each appliance.

8. Concept Box – Why domestic devices are not connected in series?

If devices are connected in series:

- The same current flows through all devices.
- If one device fails or is switched off, the circuit breaks and all devices stop working.
- Voltage across each device divides, so appliances will not get the required voltage and may not work properly.

Therefore, domestic devices are connected in parallel.

9. Series vs Parallel Combination of Resistors

Feature	Series Combination	Parallel Combination
Connection	End to end	Across the same two points
Potential difference	Divides among resistors	Same across each resistor
Current	Same through all resistors	Divides among branches; $I = I_1 + I_2 + I_3$
Equivalent resistance	$R_s = R_1 + R_2 + R_3$ (greater)	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ (smaller)
Effect of removing one resistor	Circuit breaks; all stop working	Other branches continue to work
Use	Voltage dividers, simple circuits	Domestic wiring, distribution, practical use



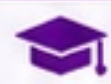
IMPORTANT TERMS

- Branch** – A separate path in a parallel circuit.
- Equivalent resistance (R_p)** – The single resistance that can replace a combination of resistors and produce the same effect.
- Current division** – The splitting of total current into branches in a parallel circuit.
- Independent operation** – Appliances working separately without affecting others.



QUICK RECALL

- Parallel – same voltage, current splits.
- $I = I_1 + I_2 + I_3$
- $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
- R_p is less than the smallest branch resistance.



KEY IDEA

In parallel combination, resistors are connected across the same two points, so every branch gets the same voltage. This creates multiple paths for current, making the circuit more efficient and reliable – which is why our homes are wired in parallel.



EXAM TIPS

- Always write the formula first.
- If resistors are more than three, extend: $\frac{1}{R_p} = \sum \frac{1}{R_i}$
- Check units (Ω for resistance, A for current, V for voltage).



CENTUM FOCUS

- Understand why parallel circuits are preferred in daily life.
- Solve mixed problems with different resistors confidently.
- Draw neat circuit diagrams and label currents and resistances.

Ch 11: Electricity

SECTION G – HEATING EFFECT OF ELECTRIC CURRENT AND JOULE’S LAW

1. Why does electric current produce heat?

- When electric current flows through a conductor, the free electrons collide with the atoms/ions of the conductor.
- These collisions transfer energy to the atoms, causing them to vibrate faster.
- This increase in internal energy of the conductor is perceived to us as heat.
- Thus, electrical energy is converted into heat energy.

2. Role of resistance – Conversion of electrical energy into heat

- Every conductor offers some resistance to the flow of current.
- Due to resistance, the moving electrons lose energy in the form of heat.
- The greater the resistance, the more is the heat produced (for the same current and time).

3. Joule’s Law of Heating

The amount of heat (H) produced in a conductor is directly proportional to the square of the current (I) passing through it, to the resistance (R) of the conductor and to the time (t) for which the current flows.

Mathematically,

$$H = I^2 R t$$

where, H = Heat produced (in joules)

I = Current (in amperes)

R = Resistance of the conductor (in ohms)

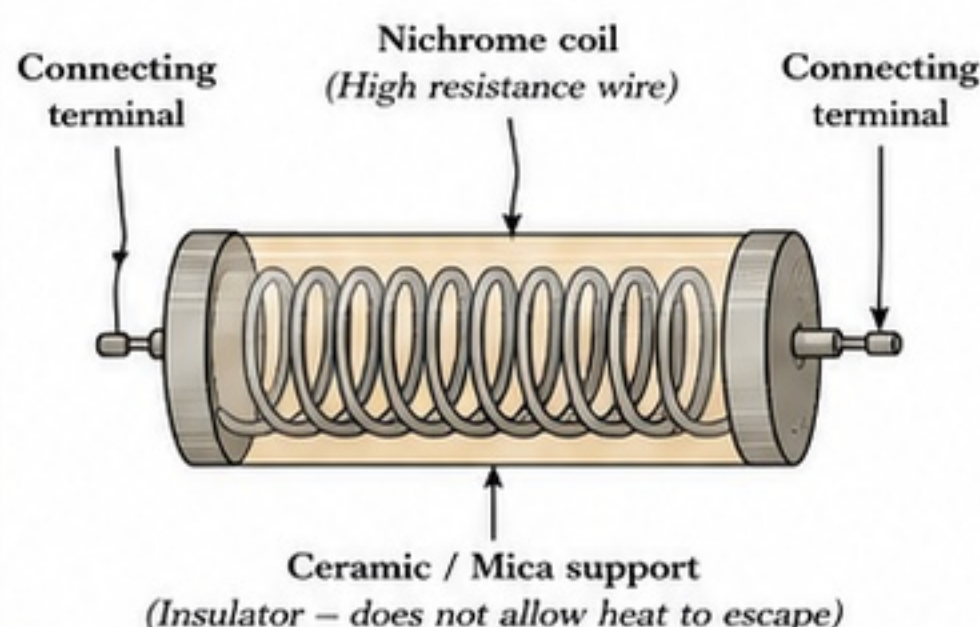
t = Time for which current flows (in seconds)

Other equivalent forms:

► Using $V = IR$, we get $H = V I t$
(Heat = Voltage × Current × Time)

► Using $I = \frac{V}{R}$, we get $H = \frac{V^2 t}{R}$

DIAGRAM – HEATING COIL (HEATING ELEMENT)



Why nichrome wire?

- High resistivity** – offers high resistance and produces more heat.
- High melting point** – can be heated to red hot without melting.
- Does not oxidize easily** – maintains its properties for long time.
- Strong and durable** – does not break easily.

4. Dependence of heating effect

- On current (I):** Heat is directly proportional to the square of current. (If current is doubled, heat becomes four times.)
- On resistance (R):** Heat is directly proportional to resistance. (If resistance is doubled, heat becomes double.)
- On time (t):** Heat is directly proportional to time. (If time is doubled, heat becomes double.)

Thus, to produce more heat, we can increase current, increase resistance or increase time.

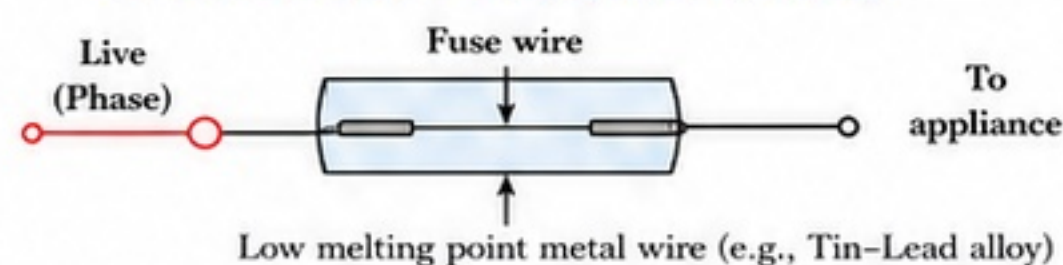
5. Practical applications of heating effect

- Electric iron** – to press clothes.
- Toaster** – to toast bread.
- Room heater** – to warm the room.
- Electric kettle** – to boil water.
- Geyser** – to heat water for bath.
- Filament bulb** – filament glows white hot and produces light.
- Fuse** – melts and breaks the circuit during overload or short circuit.

6. Useful and Unwanted Heating Effects

Useful Heating Effect (Beneficial)	Unwanted Heating Effect (Harmful)
Used in electric appliances to produce heat for our comfort.	Causes loss of electrical energy in the form of heat (wastage).
Helps in cooking, heating water, warming rooms, etc.	Overheating of wires and appliances can damage insulation and parts.
Used in devices like heater, iron, toaster, geyser, etc.	May cause short circuit, fire or damage to electrical circuits.
Makes life easier and more comfortable.	Needs to be minimized in power transmission and electrical circuits.

7. Electric Fuse – Concept and Working



- A fuse is a short piece of wire made of a low melting point alloy connected in series in the live wire of a circuit.
- If current exceeds the safe limit due to overloading or short circuit, the fuse wire gets overheated (due to Joule’s heating) and melts.
- When the fuse wire melts, the circuit breaks and stops the current.
- This protects appliances and wiring from damage and prevents fire.



QUICK RECALL

- Heat is produced due to collisions between electrons and atoms.
- Joule’s law: $H = I^2 R t$
- Also, $H = V I t$ and $H = \frac{V^2 t}{R}$
- Heat $\uparrow$ with I , R and t .
- Fuse wire melts to break the circuit and protect from danger.



KEY IDEA

- Electrical energy is converted into heat due to resistance.
- The amount of heat produced depends on current, resistance and time.
- Nichrome is used in heating elements because it has high resistivity, high melting point and does not oxidize easily.
- Fuse is a safety device that melts during excess current and breaks the circuit.



EXAM TIPS

- Always write Joule’s law as $H = I^2 R t$ and explain the dependence.
- If voltage is given, use $H = V I t$ or $H = \frac{V^2 t}{R}$ as required.
- Draw neat labelled diagrams of heating coil and fuse.
- Give at least two practical applications in answers.



CENTUM FOCUS

- Practice numerical problems using $H = I^2 R t$, $H = V I t$ and $H = \frac{V^2 t}{R}$.
- Compare useful vs unwanted heating effects.
- Draw and label diagrams correctly for full marks.



Revise smart, draw diagrams neatly, and score full marks.



Ch 11: Electricity

Part 1 of 1 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION H - ELECTRIC POWER, ELECTRICAL ENERGY AND COMMERCIAL UNIT

1. What is Electric Power?

- Electric power is the rate at which electrical work is done or the rate at which electrical energy is consumed in an electric circuit.
- If W is the electrical work done in time t , then power P is
$$P = \frac{W}{t} \quad (\text{Rate of doing electrical work})$$
- In a circuit, work done $W = VIt$ (Potential difference $\times$ Current $\times$ Time). Hence,

$$P = \frac{VIt}{t} = VI$$

2. Power in Terms of Resistance

- From Ohm's law, $V = IR$
- Substitute in $P = VI$

Using $V = IR$ in $P = VI$

$$P = I(IR) = I^2R$$

Using $I = \frac{V}{R}$ in $P = VI$

$$P = V \left(\frac{V}{R} \right) = \frac{V^2}{R}$$

3. Unit of Power

- SI unit of power is watt (W).
- 1 watt is the power when 1 joule of electrical energy is consumed in 1 second.

$$1 \text{ Watt (W)} = 1 \text{ Joule/second (J s}^{-1}\text{)}$$

- 1 kilowatt (kW) = 1000 W = 10^3 W

4. Power Rating of Appliances

- The power rating of an appliance is the amount of electric power it requires to operate under normal conditions.
- It is usually written on the body of the appliance in watts (W) or kilowatts (kW).
- Higher the power rating, greater is the rate of electrical energy consumption.

Common Domestic Appliances and Their Power Ratings

Appliance	Typical Power Rating	Typical Use
LED Bulb	8 W – 12 W	Lighting
Tube Light (LED)	18 W – 22 W	Lighting
Ceiling Fan	60 W – 80 W	Cooling
Table Fan	40 W – 60 W	Cooling
Television (LED)	80 W – 120 W	Entertainment
Refrigerator (Star Rated)	150 W – 250 W	Cooling
Washing Machine	400 W – 600 W	Washing
Electric Iron	750 W – 1000 W	Ironing
Water Heater (Geyser)	1500 W – 3000 W	Heating Water
Microwave Oven	1000 W – 1500 W	Cooking
Air Conditioner (1.5 ton)	1500 W – 2000 W	Cooling

💡 Actual power may vary with brand, model and usage conditions.

5. Electrical Energy

- Electrical energy consumed in time t by an appliance of power P is
$$E = Pt$$
where E is in joules (J),
 P in watts (W), t in seconds (s).
- If power is in kW and time is in hours, then energy is in kilowatt-hour (kWh).

$$E(\text{kWh}) = P(\text{kW}) \times t(\text{h})$$

6. Commercial Unit of Electrical Energy

- The commercial unit of electrical energy is kilowatt-hour (kWh).
- 1 kWh is the electrical energy consumed when a power of 1 kW is used for 1 hour.

$$1 \text{ kWh} = 1 \text{ kW} \times 1 \text{ h} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$$

7. Power vs Energy – Key Distinction

Feature	Power (Watt)	Energy (Kilowatt-hour)
Meaning	Rate of doing work or consumption of energy.	Total electrical energy consumed over time.
Formula	$P = VI = I^2R = \frac{V^2}{R}$	$E = Pt$ (in joules) $E(\text{kWh}) = P(\text{kW}) \times t(\text{h})$
Unit (SI/Commercial)	Watt (W), kilowatt (kW)	Kilowatt-hour (kWh)
Depends on	Voltage, current, resistance	Power and time
Analogy	Speed of a car	Distance travelled

8. Electricity Bill – Simple Concept

- Electricity supplied to homes is measured in units (kWh) by an energy meter.
- Your monthly bill = (Units consumed $\times$ rate per unit) + fixed charges + taxes.
- Rate per unit (₹/kWh) varies with the electricity board and slab (if applicable).
- Reducing power usage and usage time lowers the electricity bill.

Example Bill Structure (Illustrative)

Particulars	Amount (₹)
Units consumed	150 kWh
Rate per unit	₹ 8.00 / kWh
Energy charges	$150 \times 8.00 = ₹ 1200.00$
Fixed charges	₹ 100.00
Taxes (e.g., 10%)	₹ 130.00
Total Amount Payable	₹ 1430.00

💡 Note: Rates and taxes differ across regions and slabs.

9. Solved Numerical Frameworks

(A) Energy Consumption in kWh

Framework: $E(\text{kWh}) = P(\text{kW}) \times t(\text{h})$

Example: A 1500 W heater is used for 2.5 hours daily. Find energy consumed in 30 days in kWh.

Solution:

- $P = 1500 \text{ W} = 1.5 \text{ kW}$
- Daily energy = $1.5 \times 2.5 = 3.75 \text{ kWh}$
- Monthly (30 days) = $3.75 \times 30 = 112.5 \text{ kWh}$

Answer: 112.5 kWh

(B) Electricity Bill Calculation

Framework:

Total Bill = (Units $\times$ Rate per unit) + Fixed Charges + Taxes

Example: A TV (100 W) is used 5 hours/day and a fan (70 W) 8 hours/day for 30 days. Rate = ₹ 8 per unit, Fixed charges = ₹ 100, Tax = 10%. Find the bill.

Solution:

- TV power = 0.1 kW $\rightarrow$ Energy = $0.1 \times 5 \times 30 = 15 \text{ kWh}$
- Fan power = 0.07 kW $\rightarrow$ Energy = $0.07 \times 8 \times 30 = 16.8 \text{ kWh}$
- Total units = $15 + 16.8 = 31.8 \text{ kWh}$
- Energy charges = $31.8 \times 8 = ₹ 254.40$
- Total bill = $(254.40 + 100) + 10\% \text{ of } 254.40 = 354.84$

Answer: ₹ 354.84

Conversion Corner

- 1 h = 3600 s
- 1 kW = 1000 W
- 1 kWh = $3.6 \times 10^6 \text{ J}$
- Energy (J) = Power (W) $\times$ Time (s)
- Energy (kWh) = Power (kW) $\times$ Time (h)

QUICK RECALL

- $P = VI = I^2R = \frac{V^2}{R}$
- Unit of power: watt (W)
- $E = Pt$
- 1 kWh = $3.6 \times 10^6 \text{ J}$

KEY IDEA

Power tells how fast energy is used.
Energy tells how much is used.
Multiply power by time to get energy, and convert to kWh for billing and comparisons.

EXAM TIPS

- Learn $P = VI$, $P = I^2R$, $P = \frac{V^2}{R}$
- Remember $E = Pt$ and unit conversions.
- In numericals, convert W to kW and seconds to hours properly.
- Watch units: W (power) vs kWh (energy).

CENTUM FOCUS

- Learn common appliance ratings.
- Practise energy/bill problems daily.
- Distinguish clearly between power and energy.
- Revise formulae and conversions.



Revise smart, draw diagrams neatly, and score full marks.



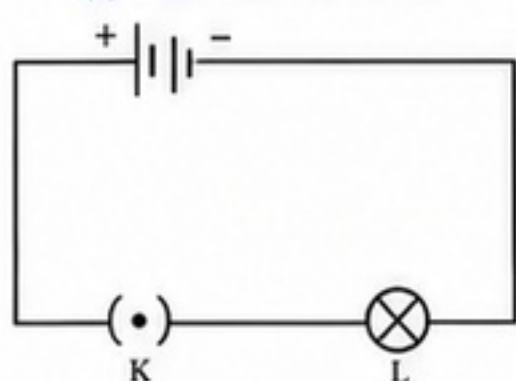
Ch 11: Electricity

Part 1 of 1 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

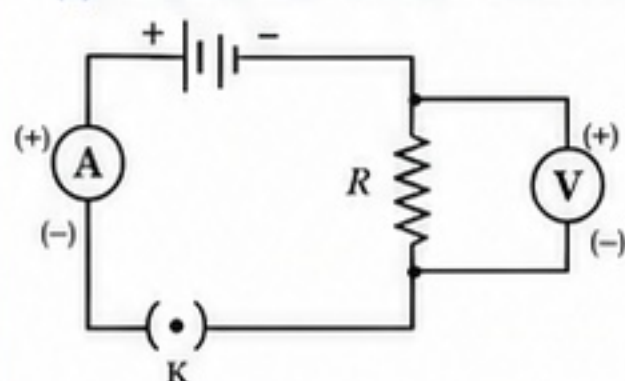
SECTION I - IMPORTANT DIAGRAMS, FORMULA FRAMES, NUMERICAL TIPS AND HIGH-VALUE QUESTIONS

1. MUST-KNOW DIAGRAMS

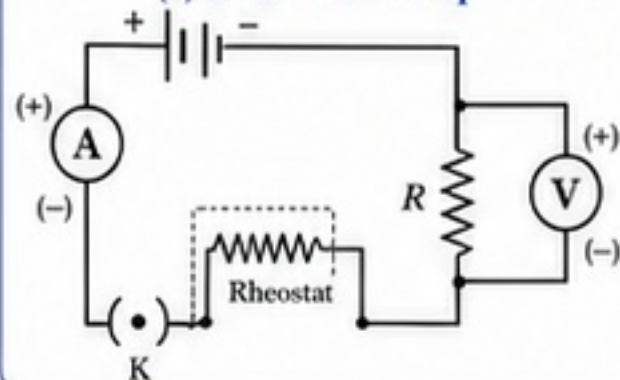
(a) Basic Electric Circuit



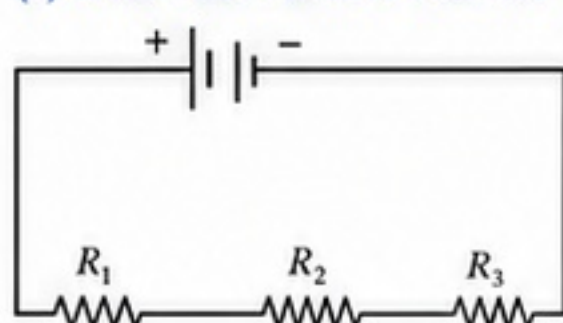
(b) Ammeter and Voltmeter Connection



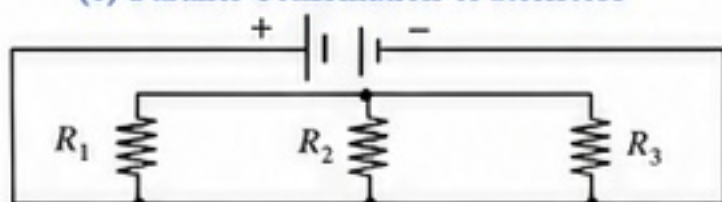
(c) Ohm's Law Setup



(d) Series Combination of Resistors



(e) Parallel Combination of Resistors



2. FORMULA SHEET (Formula Frame)

No.	Formula	Where,
1.	$I = \frac{Q}{t}$	I = current (A), Q = charge (C), t = time (s)
2.	$V = \frac{W}{Q}$	V = potential difference (V), W = work (J), Q = charge (C)
3.	$V = IR$	V = potential difference (V), I = current (A), R = resistance (Ω)
4.	$R = \frac{\rho l}{A}$	R = resistance (Ω), ρ = resistivity ($\Omega \cdot \text{m}$), l = length (m), A = area of cross-section (m^2)
5.	$R_s = R_1 + R_2 + \dots$	R_s = equivalent resistance in series
6.	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$	R_p = equivalent resistance in parallel
7.	$H = I^2 R t$	H = heat produced (J), I = current (A), R = resistance (Ω), t = time (s)
8.	$P = VI$	P = power (W), V = potential difference (V), I = current (A)
9.	$E = Pt$	E = energy (J or kWh), P = power (W), t = time (s or h)
10.	$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$	Energy unit conversion

3. BOARD-STYLE SHORT ANSWER FRAMES

Question	Answer Frame (Use these key points)
Q1. State Ohm's law.	At constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends.
Q2. Define resistance.	Resistance is the property of a conductor that opposes the flow of electric current. $R = V/I$.
Q3. Define resistivity.	Resistivity (ρ) is the resistance of a conductor of unit length and unit cross-sectional area. $\rho = RA/l$.
Q4. Why are domestic circuits connected in parallel?	- Each appliance gets the same potential difference (230 V). - Appliances work independently. - Failure of one appliance does not affect others.
Q5. Why are heating elements made of alloys?	- Alloys have high resistivity $\rightarrow$ produce more heat. - High melting point $\rightarrow$ do not melt at high temperature. - Resist oxidation $\rightarrow$ last longer.
Q6. What is 1 kWh?	1 kWh is the energy consumed when a power of 1 kW is used for 1 hour. $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

4. NUMERICAL TIPS (Quick Steps)

- ☒ **Step 1:** Read the question carefully and note the given values and unit.
- ☒ **Step 2:** Convert all given values to SI units.

Common Conversions

Quantity	To SI Unit	Conversion
Length	m	$1 \text{ cm} = 10^{-2} \text{ m}$, $1 \text{ mm} = 10^{-3} \text{ m}$
Area	m^2	$1 \text{ cm}^2 = 10^{-4} \text{ m}^2$, $1 \text{ mm}^2 = 10^{-6} \text{ m}^2$
Resistance	Ω	$1 \text{ k}\Omega = 10^3 \Omega$, $1 \text{ M}\Omega = 10^6 \Omega$
Energy	J	$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$
Time	s	$1 \text{ min} = 60 \text{ s}$, $1 \text{ h} = 3600 \text{ s}$

- ☒ **Step 3:** Write the correct formula from the list.
- ☒ **Step 4:** Substitute values with units, check consistency.
- ☒ **Step 5:** Calculate carefully and keep proper significant figures.
- ☒ **Step 6:** Write the answer with unit and box it.

5. COMMON MISTAKES TO AVOID

Mistake	Why It's Wrong / Correction
Forgetting units in formula	Always write units with values.
Using $V = IR$ when R changes	Use only when temperature is constant.
Wrong connection of meters	Ammeter in series, voltmeter in parallel.
Adding resistances in parallel	Use $1/R_p = 1/R_1 + 1/R_2 + \dots$, not $R_p = R_1 + R_2 + \dots$
Confusing current and voltage	Current (I) flows in a circuit, voltage (V) acts across.
Ignoring power ratings	Do not exceed appliance power; may burn out.
Wrong area in $R = \rho l/A$	Use cross-sectional area in m^2 , not radius or diameter.
Forgetting conversion	Convert every value to SI unit before solving.

6. HIGH-VALUE QUESTIONS (Practice Set)

- State and explain Ohm's law with a suitable circuit diagram.
- Derive the expression for the resistance of a conductor in terms of its length, area of cross-section and resistivity.
- Three resistors of 2Ω , 3Ω and 6Ω are connected (a) in series and (b) in parallel to a 9 V battery. Calculate the current in each case.
- A heater of resistance 10Ω is connected to a 220 V source. Calculate (a) current, (b) power, (c) heat produced in 5 minutes.
- Explain why series combination of resistors is not used in domestic wiring.
- Calculate the cost of energy consumed by a 1000 W appliance that is used daily for 4 hours for 30 days. The cost of 1 kWh is ₹ 7.



QUICK RECALL

- Current = flow of charge.
- $V = IR$ (Ohm's law).
- Higher resistance $\rightarrow$ less current.
- Series $\rightarrow$ same current, voltages divide.
- Parallel $\rightarrow$ same voltage, currents divide.



KEY IDEA

Electricity is understood through the relationship between current, voltage, resistance and power. Master the connections, use correct formulas, convert units and apply them step-by-step to solve any numerical confidently.



EXAM TIPS

- Always draw neat circuit diagrams.
- Label meters and components.
- Write formula and units clearly.
- Show all steps in numericals.
- Check units and reasonableness of the final answer.



CENTUM FOCUS

- Practise previous year questions.
- Solve numericals daily.
- Create your own formula sheet.
- Revise diagrams, formulae and unit conversions regularly.
- Attempt full-length tests.

Ch 11: Electricity

SECTION J - MASTER REVISION SHEET, NCERT TRAPS AND CENTUM FOCUS

1. COMPACT DEFINITIONS

Term	Definition
Current (I)	Rate of flow of electric charge through a conductor. $I = \frac{Q}{t}$; Unit: ampere (A).
Potential Difference (V)	Work done per unit charge in moving a charge between two points. $V = \frac{W}{Q}$; Unit: volt (V).
Resistance (R)	Opposition offered by a conductor to the flow of current. $R = \frac{V}{I}$; Unit: ohm (Ω).
Resistivity (ρ)	Intrinsic property of a material that resists the flow of current. $\rho = \frac{RA}{l}$; Unit: ohm metre ($\Omega \text{ m}$).
Series Combination	Resistors connected end to end; same current flows through each. $R_S = R_1 + R_2 + R_3 + \dots$
Parallel Combination	Resistors connected across the same two points; same voltage across each. $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$
Joule Heating	Heat produced in a conductor due to the flow of current. $H = I^2 R t$; Unit: joule (J).
Electric Power (P)	Rate at which electrical energy is consumed or work is done. $P = VI = I^2 R = \frac{V^2}{R}$; Unit: watt (W).
Electric Energy (E)	Total electrical energy consumed. $E = Pt = VIt = I^2 R t = \frac{V^2 t}{R}$; Unit: joule (J).
Kilowatt-hour (kWh)	Commercial unit of electrical energy. $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$; Used in electricity bills.

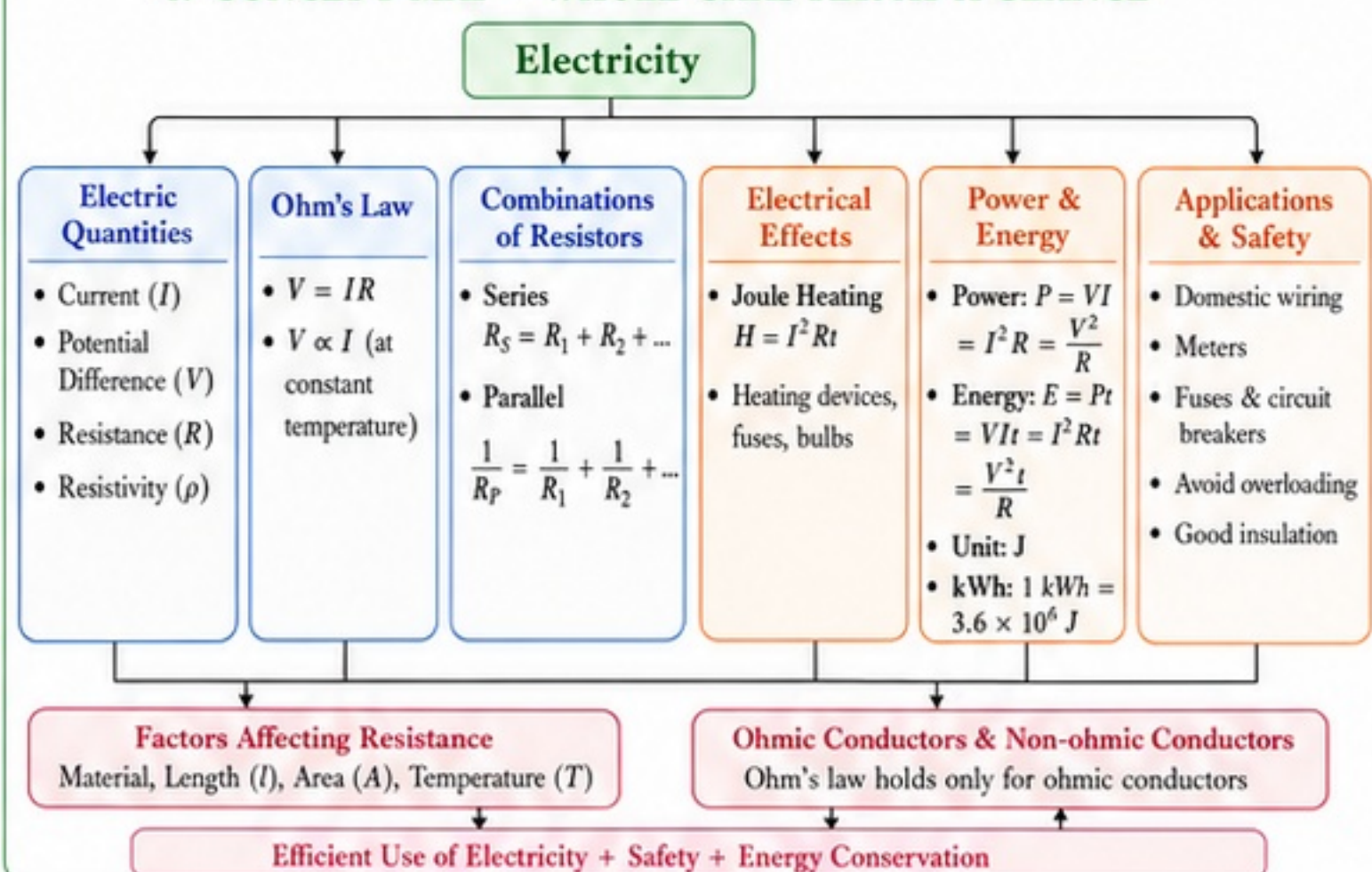
2. MASTER SUMMARY – THE WHOLE CHAPTER IN ONE VIEW

- **Electric Current:** Flow of charge; $I = Q/t$; Unit: ampere (A).
- **Potential Difference:** $V = W/Q$; causes current to flow.
- **Ohm's Law:** At constant temperature, $V \propto I$; $V = IR$.
- **Resistance:** $R = V/I$; Unit: ohm (Ω). Depends on material, length (l), area (A) and temperature.
- **Resistivity:** $\rho = \frac{RA}{l}$; lower $\rho \Rightarrow$ better conductor.
- **Series Combination:** Same current; $R_S = R_1 + R_2 + \dots$; $V_T = V_1 + V_2 + \dots$
- **Parallel Combination:** Same voltage; $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$; $I_T = I_1 + I_2 + \dots$
- **Joule Heating:** $H = I^2 R t$; depends on I^2 , R and t .
- **Electric Power:** $P = VI = I^2 R = \frac{V^2}{R}$; Unit: watt (W).
- **Electric Energy:** $E = Pt = VIt = I^2 R t = \frac{V^2 t}{R}$; Unit: joule (J).
- **Kilowatt-hour:** $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$; used in electricity billing.
- **Practical Applications:** Heating devices, fuses, bulbs, resistors, wiring, meters.
- **Safety:** Use correct fuse rating, avoid overloading, ensure good insulation.
- **Key Relationships:** Double I (at same R) $\rightarrow P$ becomes 4 times.
Double V (at same R) $\rightarrow P$ becomes 4 times.
- **Important Factors:** Material, length, area of cross-section and temperature determine resistance.

3. NCERT TRAPS AND MISCONCEPTIONS

Trap / Misconception	Correct Explanation
Current and Voltage are same.	Current is flow of charge; voltage is potential difference that causes the current.
Resistance and Resistivity are same.	Resistance (R) depends on size and shape; resistivity (ρ) is an intrinsic property of material.
Power and Energy are same.	Power is rate of doing work (Watt); energy is total work done (Joule).
Series vs Parallel confusion.	Series: same current, voltages add. Parallel: same voltage, currents add.
Watt and Unit (kWh) are same.	Watt is power; Unit (kWh) is energy consumed in 1 hour.
Higher current always means better device performance.	Depends on design; excess current may overheat and damage the device.
Ohm's law does not apply always.	Ohm's law holds only if temperature remains constant and physical conditions of conductor do not change.
More bulbs in series gives more brightness.	In series, current decreases; bulbs glow dimmer.
More bulbs in parallel consumes less energy.	In parallel, current divides; total power increases.
Zero resistance means infinite current.	Not possible in real circuits as source, internal resistance and safety fuses limit current.

4. CONCEPT MAP – WHOLE CHAPTER AT A GLANCE



5. 10 MUST REMEMBER

1. $I = Q/t$ (Current).
2. $V = W/Q$ (Potential Difference).
3. $V = IR$ (Ohm's Law).
4. $R = \rho l/A$ (Resistance in terms of resistivity).
5. Series: $R_S = R_1 + R_2 + \dots$
6. Parallel: $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$
7. Joule Heating: $H = I^2 R t$.
8. Power: $P = VI = I^2 R = \frac{V^2}{R}$.
9. Energy: $E = Pt = VIt = I^2 R t = \frac{V^2 t}{R}$.
10. $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

6. FINAL CENTUM CHECKLIST ✓

- ☐ I can define current, voltage, resistance and resistivity.
- ☐ I can apply Ohm's Law problems confidently.
- ☐ I can solve series and parallel resistor problems.
- ☐ I can calculate Joule heating in real-life cases.
- ☐ I can find power and energy using all formula forms.
- ☐ I can convert units (W, J, kWh) correctly.
- ☐ I can solve numerical problems with correct units.
- ☐ I can differentiate between resistance and resistivity.
- ☐ I can explain factors affecting resistance.
- ☐ I can identify and correct common misconceptions.

7. BEFORE EXAM – QUICK GLANCE

- 👁 Know all definitions and formulas.
- ⚡ Differentiate: Current vs Voltage, R vs ρ , P vs E .
- 🔍 Remember series vs parallel rules.
- ✳️ Learn Joule heating, power and energy formulas.
- 🕒 Revise units: A, V, Ω , W, J, kWh.
- 📝 Solve past year numericals and diagrams.
- 🔍 Check NCERT in-text and end questions once more.



Revise smart, use formulas carefully, and score full marks.

